

UNIVERSITY COLLEGE OF ENGINEERING

Examination

MATHEMATICS-IV (KAS302/KAS402)

Roll No. _____

Subject: MATHEMATICS-IV

Time: 3 Hours

Max. Marks: 70

Code: KAS302/KAS402

General Instructions:

1. Attempt all questions.
2. Each question carries marks as indicated in the respective sections.
3. The use of scientific calculators is permitted.

SECTION A — SECTION A

Attempt all questions

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	Define Lagrange's linear partial differential equation and write its standard form.	2	CO1
2	Classify the second-order partial differential equation: $\partial^2 u / \partial x^2 + 4 \partial^2 u / \partial x \partial y + 4 \partial^2 u / \partial y^2 = 0$.	2	CO2
3	Define the Moment Generating Function (MGF) of a random variable.	2	CO3
4	List any two properties of regression coefficients.	2	CO3
5	State the addition law of probability for two non-mutually exclusive events.	2	CO4
6	Distinguish between Null Hypothesis and Alternative Hypothesis.	2	CO5
7	Define the term 'Level of Significance' as used in hypothesis testing.	2	CO5

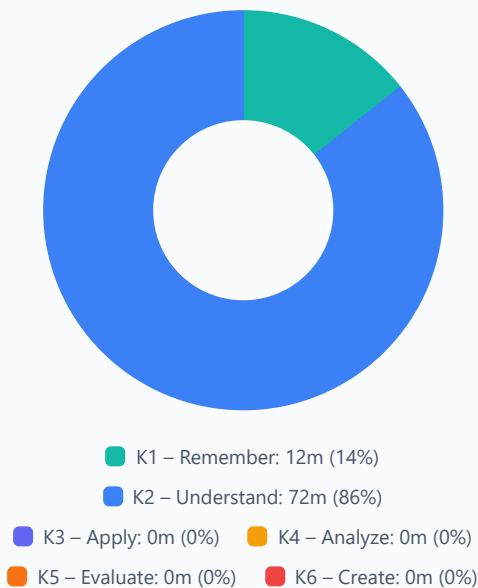
SECTION B — SECTION B*Attempt all questions**7 marks each | Attempt all questions*

Q.No.	Question	Marks	CO
8	Explain Charpit's method for finding the complete integral of a non-linear partial differential equation of first order.	7	CO1
9	Describe the method of separation of variables for solving a second-order partial differential equation.	7	CO2
10	Explain the principle of least squares and its application in fitting a straight line to a given set of data points.	7	CO3
11	State and explain Bayes' theorem for the probability of causes with a suitable mathematical expression.	7	CO4
12	Discuss the procedure for testing a hypothesis concerning the difference between two means for small samples using the T-test.	7	CO5

SECTION C — SECTION C			
<i>Attempt all questions. For each question, choose either part (a) or part (b).</i>			
<i>7 marks each Attempt all questions</i>			
Q.No.	Question	Marks	CO
13	(a) Compare linear and non-linear partial differential equations of the first order and explain the origin of PDEs.	7	CO1
	<i>OR</i>		
	(b) Explain the Cauchy's method of characteristics for solving first-order partial differential equations.	7	CO1
14	(a) Discuss the general solution of the one-dimensional wave equation using the method of separation of variables.	7	CO2
	<i>OR</i>		
	(b) Explain the Laplace equation in two dimensions and its application in steady-state heat flow.	7	CO2
15	(a) Define Skewness and Kurtosis. Explain how they describe the shape of a frequency distribution.	7	CO3
	<i>OR</i>		
	(b) Explain the concept of regression analysis and why there are two lines of regression.	7	CO3
16	(a) Describe the Normal distribution and list its key properties and its probability density function.	7	CO4
	<i>OR</i>		
	(b) Explain the Poisson distribution as a limiting case of the Binomial distribution and state its mean and variance.	7	CO4
17	(a) Explain the concept of Statistical Quality Control (SQC) and the role of control charts in monitoring processes.	7	CO5
	<i>OR</i>		
	(b) Describe the construction and interpretation of X-bar and R charts for variables.	7	CO5

Question Paper Analysis

Bloom's Taxonomy Distribution



Course Outcome Distribution

